AqSol-Blend: A CPU-Trained Descriptor Ensemble for Aqueous Solubility Prediction

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Abstract

AqSol-Blend investigates molecular representation changes within a reproducible CPU-trained ensemble for aqueous solubility prediction. The study combines controlled development comparisons, an overlap-screened external evaluation and fresh-environment training reproduction. The method integrates LightGBM, Mordred-based CatBoost and robust-scaled support-vector regression with fixed weights of 0.675, 0.225 and 0.100. On the Therapeutics Data Commons AqSolDB benchmark, the ensemble achieves a mean absolute error of 0.736 ± 0.014 across five runs. Substituting Mordred descriptors for the CatBoost branch's 2,937 fingerprint and RDKit features reduces development MAE by 0.0078 and improves nine of ten splits. In contrast, seed averaging, alternative CatBoost losses and reduced LightGBM capacity do not improve the development baseline. On an overlap-screened external cohort of 360 compounds from BigSolDB, the frozen five-run average obtains MAE 1.008, compared with 1.017 for its predecessor and 1.065 for a fixed random-forest baseline. The external improvement over the predecessor remains uncertain. Development reused validation partitions and included four sequential benchmark test evaluations; external results are interpreted separately because measurement conditions and concentration conventions differ.

1 Introduction and related work

Aqueous solubility prediction requires a representation of molecular structure that is informative for the target and a regression model that can use it effectively. Fingerprints describe structural patterns, whereas molecular descriptors summarize a broader range of graph and physicochemical properties. These representations need not yield the same predictive errors, motivating their combination within an ensemble.

AqSolDB provides a curated collection of solubility measurements and molecular descriptors [2]; Therapeutics Data Commons (TDC) supplies a standardized benchmark split and evaluation interface [3]. Previous work has demonstrated the value of combining molecular representations. MapLight uses boosted trees with multiple fingerprints and descriptors [4], while MiniMol learns transferable fingerprints through large-scale pretraining [5].

AqSol-Blend examines a narrower question: can a fixed combination of conventional regressors benefit from assigning different molecular representations to its branches? The method combines a fingerprint-based LightGBM predictor, a Mordred-based CatBoost predictor and descriptor-based support-vector regression. Training uses no external solubility labels or pretrained representations. The experiments compare representation changes with changes to training objectives, model capacity and seed averaging, documenting both successful and unsuccessful alternatives.

MapLight also evaluates learned graph-network fingerprints alongside handcrafted features [4]. The TDC solubility leaderboard lists Chemprop-RDKit at 0.761 ± 0.025, AttentiveFP at 0.776 ± 0.008, and MapLight+GNN at 0.789 ± 0.003 [12]. These entries place the present descriptor ensemble alongside graph-based and hybrid approaches, while MiniMol represents pretrained molecular representation learning. Published leaderboard values are contextual comparisons rather than paired experiments performed here.

For measurement context, Palmer and Mitchell estimated uncertainty of 0.6–0.7 log solubility units in their literature-derived druglike-molecule dataset, compared with typically less than 0.05 for their CheqSol measurements [11]. Those estimates are not a measured uncertainty for AqSolDB and do not determine the significance of differences between models evaluated on the same labels.

Mordred [6] supplies the final CatBoost features. A public solubility notebook motivated this substitution [7]; the present experiments use a finite growth-policy comparison rather than reproduce that notebook's complete feature-selection and optimization procedure.

2 Data and evaluation protocol

Experiments used the TDC AqSolDB development pool of 7,985 rows and its specified test set of 1,997 rows. Targets are aqueous solubility in log mol/L. The primary error measure is

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|. \tag{1}$$

Final per-run models use the default TDC scaffold development splits with seeds 1–5. Additional development comparisons used seeds 17, 29, 43, 71 and 101. The same development partitions were used for comparisons across model variants.

The reported benchmark result is the mean and population standard deviation returned by `evaluate_many` for five per-seed ensembles. TDC rounds the constituent metrics and aggregate outputs to three decimal places. The average prediction across the five runs is used for the separate external evaluation described below; it has not been scored as a separate TDC benchmark entry.

Tree-model training length was selected using the corresponding development validation partition. Separate inner-stopping experiments are described below for the predecessor ensemble.

2.1 External transfer protocol

The external cohort was assembled from BigSolDB version 2.2, archived under DOI 10.5281/zenodo.22648301 [13], with the dataset methodology described by Krasnov et al. [1]. Eligibility was restricted to water measurements at 297.15–299.15 K. Molecular identities were compared against all 9,982 AqSolDB rows after parent-fragment selection, charge neutralization, tautomer canonicalization and removal of stereochemistry. For one AqSolDB structure whose tautomer standardization failed, all external candidates with the same parent heavy-element composition were conservatively excluded. Candidates with binary Morgan fingerprint similarity of 1.0 to any development molecule were also excluded. This yielded 421 measurement rows representing 361 standardized parent identities; one row without a finite target was excluded at scoring, leaving 360 identities.

The protocol, cohort and model choices were recorded before computing external errors. Predictions from both frozen ensembles were averaged across their five runs. A fixed random forest with 500 trees, minimum leaf size 2 and all features available at each split was trained on the 210 RDKit descriptors of the full AqSolDB development pool (seed 20261026). No external labels were used in training or model selection. All predictions were saved before loading external targets for error calculation.

Targets were aggregated by taking the median within each parent-identity/source pair, then the median across sources. Predictions for unique input SMILES were averaged within each parent identity. MAE was the primary metric, with RMSE secondary. Paired differences were summarized using 2,000 molecule-level bootstrap resamples (seed 20261026). The supplied `LogS(mol/L)` values use moles per litre of solvent, approximating molarity in dilute solutions [13]; this distinction is retained in interpreting the results.

3 Method

For each seed, the final prediction is

$$\hat{y} = 0.675 f_{\mathrm{LGB}}(x) + 0.225 f_{\mathrm{Cat}}(x) + 0.100 f_{\mathrm{SVR}}(x). \tag{2}$$

The mixture preserves the original LightGBM/CatBoost ratio of 0.75/0.25 within a combined weight of 0.90 and assigns the remaining 0.10 to SVR. The 10% SVR contribution was evaluated in development comparisons before the Mordred substitution. Weights were deliberately held fixed during that substitution, so its comparison does not combine a representation change with weight re-optimization. Each branch uses its own molecular representation.

LightGBM branch. Inputs comprise 2,048 radius-two Morgan count features, 167 MACCS keys, 512 Avalon bits, 210 RDKit descriptors and 315 ErG features, totaling 3,252 columns. ErG uses fuzz increment 0.3 and paths of length 1–15. LightGBM [8] uses absolute-error regression, at most 2,500 trees, learning rate 0.025, 63 leaves, minimum child samples 10, L2 regularization 5, feature fraction 0.8 and row fraction 0.9 with sampling frequency 1. Deterministic column-wise fitting uses six CPU threads.

CatBoost branch. The preceding CatBoost branch used 2,937 columns: 2,048 radius-two Morgan counts, 167 MACCS keys, 512 Avalon bits and 210 RDKit descriptors, without ErG features. The final substitution replaces that entire input with Mordred descriptors while retaining the other branches and mixture weights. The community-maintained `mordredcommunity` 2.0.7 package [14], derived from the original Mordred calculator, supplies 1,612 two-dimensional descriptors after excluding `FilterItLogS`. No learned feature selection is applied. Failed, nonfinite or float32-overflow descriptor values become zero. CatBoost [9] uses SymmetricTree growth, up to 2,200 iterations, depth 8, learning rate 0.035 and L2 leaf regularization 5. Training minimizes RMSE while early stopping monitors MAE. Both tree branches use patience of 150 rounds.

SVR branch. The 210 RDKit descriptors are transformed using a RobustScaler fitted only on the training partition, with quantiles 10 and 90. Transformed values are clipped to $[-10, 10]$. RBF support-vector regression [10] uses $C = 10$, $\epsilon = 0.1$ and `gamma=scale`. There is no early stopping in this branch.

4 Development experiments

The first frozen solubility result was followed by an improved LightGBM/CatBoost blend with weights 0.75/0.25. Subsequent development examined loss functions, seed averaging, regularization, molecular representations and complementary predictors. Table 1 summarizes selected bounded comparisons against that two-model baseline on the additional five development splits. The reference two-model blend has mean development MAE 0.7851 across these five splits. Positive gains denote lower MAE.

Change to the two-model baseline	Mean MAE gain	Split wins
Three-seed LightGBM averaging	−0.0016	1/5
CatBoost MAE objective	−0.0030	1/5
CatBoost Huber objective ($\delta = 1$)	−0.0002	2/5
LightGBM minimum child samples 30	−0.0087	0/5
LightGBM 31 leaves	−0.0028	2/5
Add ErG to LightGBM	+0.0016	3/5
Main-fragment features	+0.0005	3/5
Descriptor-only LightGBM	−0.0048	1/5
LightGBM without Morgan features	−0.0037	3/5
Mix 10% robust-scaled SVR	+0.0053	4/5

Table 1: Selected development comparisons against the same two-model baseline; positive gain indicates lower MAE.

Neighbor-based and standard-scaled SVR comparisons were also examined. Validation errors were larger for molecules with low similarity to the training partition, identifying chemical novelty as a useful direction for further evaluation. Adding robust-scaled SVR and ErG together produced the preceding three-model recipe. A separate matched inner-stopping check of that predecessor improved all five splits by a mean 0.0056 MAE. This check preceded the Mordred experiment.

The final experiment replaced only the CatBoost representation, retaining LightGBM, SVR and mixture weights. Two fixed CatBoost growth policies were compared on the additional splits. SymmetricTree and Depthwise improved mean MAE by 0.0092 and 0.0062 respectively, each winning all five splits. SymmetricTree was selected using the recorded development criterion and applied unchanged to seeds 1–5. It improved four of those five splits by a mean 0.0064.

5 Results

Across ten reused splits, the final substitution reduced mean MAE by 0.0078 and improved nine splits. The single deterioration was approximately 0.0022. Post hoc analysis found positive average improvements in three fixed target ranges: below −4 (0.0213), $[-4, -2]$ (0.0070), and above −2 (0.0027). These subgroup findings use repeated validation appearances and are descriptive.

Chronological official-evaluator attempt	MAE ± SD
Original frozen solubility model	0.770 ± 0.008
Improved LightGBM/CatBoost blend	0.753 ± 0.012
ErG-LightGBM, original CatBoost, robust SVR	0.742 ± 0.014
Final Mordred CatBoost substitution	$\mathbf{0.736 \pm 0.014}$

Table 2: All four recorded official evaluations. The sequence records successive model versions rather than controlled component ablations.

The displayed final mean is numerically below the MiniMol value of 0.741 ± 0.013 on the TDC leaderboard inspected on 27 October 2026 [12]. The difference in displayed mean MAE is 0.005. The comparison uses published results; AqSol-Blend has not yet been accepted onto the leaderboard.

5.1 Branch contribution and weight sensitivity

Table 3 compares the saved component predictions on the ten development splits. For two-branch combinations, retained weights were renormalized to sum to one; no weights were fitted in this analysis. The full ensemble has the lowest mean MAE, but its advantage over the two-tree combination is uneven: SVR improves five splits, with much of the mean gain arising from two splits. The result supports an average contribution on these partitions, rather than uniform improvement.

Retained branches	Mean development MAE
LightGBM	0.7939
Mordred CatBoost	0.7859
SVR	0.8850
LightGBM + CatBoost	0.7808
LightGBM + SVR	0.7873
CatBoost + SVR	0.7839
All three	0.7775

Table 3: Post-selection branch ablation on ten reused development splits.

CatBoost alone outperforms LightGBM alone despite receiving the smaller inherited weight. A descriptive sensitivity check fixed SVR at 0.10 and assigned LightGBM weights of 0.45, 0.55, 0.675 and 0.75, with CatBoost receiving the balance. Mean development MAEs were 0.7732, 0.7743, 0.7775 and 0.7803, respectively. The inherited mixture is therefore not optimal on these reused partitions. It was retained unchanged for external evaluation.

5.2 External transfer results

Model	MAE	RMSE
Current ensemble, five-run average	1.008	1.318
Preceding ensemble, five-run average	1.017	1.335
Fixed random forest	1.065	1.415

Table 4: External transfer evaluation on 360 standardized compounds.

The current ensemble improves MAE over its predecessor by 0.0092, with a paired bootstrap 95% interval of $[-0.0012, 0.0191]$. The interval includes zero, so this comparison does not resolve the incremental benefit of the final representation change. Against the fixed random forest, the MAE reduction is 0.0577 with interval $[0.0220, 0.0908]$. This supports an advantage over that baseline under the molecule-bootstrap analysis.

For nearest-development Morgan similarities below 0.3, from 0.3 to below 0.6, and at least 0.6, cohort sizes were 69, 210 and 81; current-model MAEs were 1.243, 0.964 and 0.922. Higher error on the low-similarity group

is consistent with reduced reliability away from the development chemical space. No model settings were changed after these results.

The fixed random forest has corresponding subgroup MAEs of 1.291, 1.044 and 0.927. Thus the ensemble's mean advantage is 0.049, 0.081 and 0.006 across the three groups. Post hoc paired molecule-bootstrap intervals (2,000 resamples, seed 20261027) are $[-0.040, 0.142]$, $[0.035, 0.130]$ and $[-0.053, 0.066]$, respectively. The mean advantage extends to the 69 least-similar compounds, but that subgroup's interval includes zero. These exploratory intervals are not adjusted for multiple comparisons.

As a post hoc dataset-shift reference, the same fixed random forest was evaluated once on the 1,997-row TDC test set, obtaining MAE 0.821 and RMSE 1.121. Its external MAE of 1.065 is also higher, indicating that increased external error is not unique to AqSol-Blend. This baseline was trained once on all 7,985 development rows; it is not a five-seed TDC submission or a matched training-budget comparison. No ensemble predictions or model settings were changed for this check.

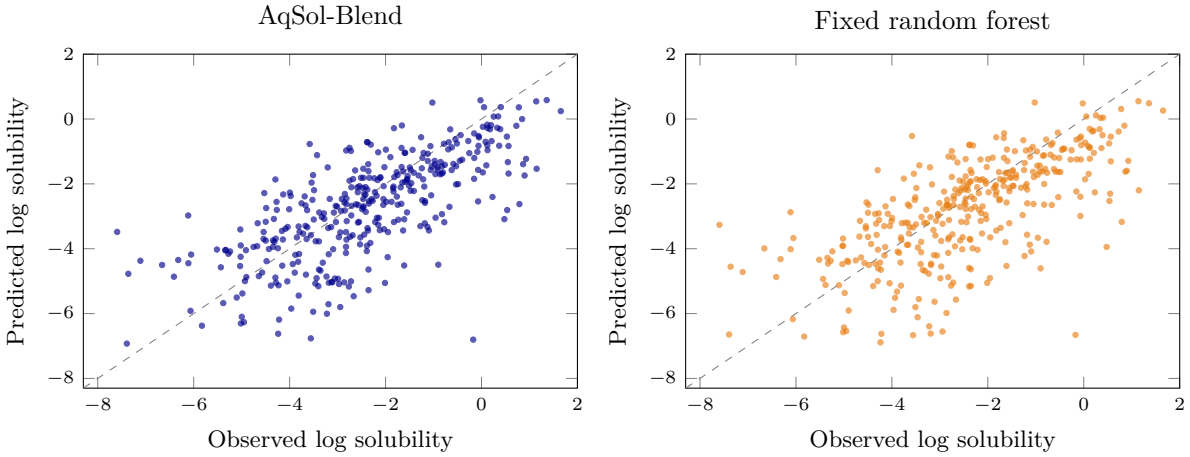


Figure 1: Observed and predicted external log solubility for 360 standardized compounds. Both panels share axes; the dashed line denotes exact agreement. Targets use the solvent-volume convention described in the protocol.

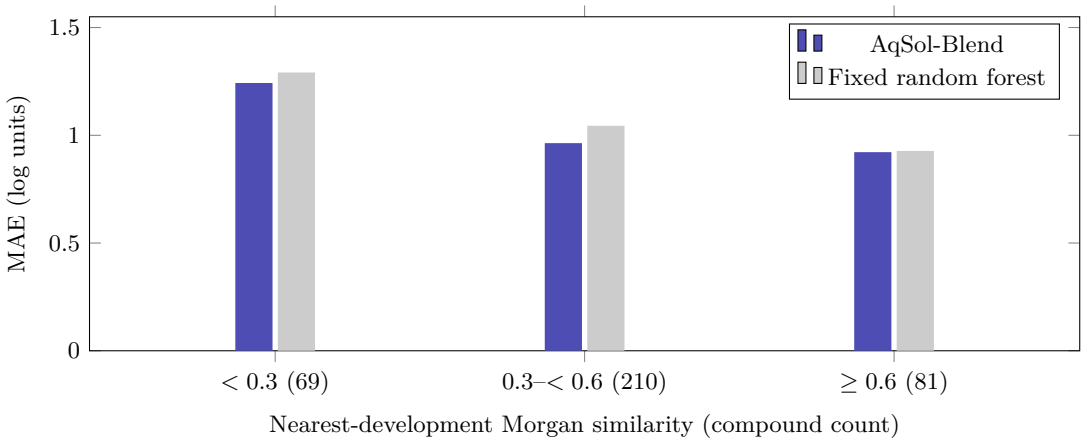


Figure 2: External MAE by nearest-development Morgan similarity. Counts refer to standardized compounds. Bars show descriptive means; paired uncertainty is reported in the text.

6 Reproducibility and limitations

6.1 Implementation and availability

The implementation retains fixed configurations, dependency versions, serialized models and ordered prediction artifacts. All five seeds were retrained in a fresh Python 3.11 environment on the original macOS ARM64 host. After verifying molecular order and target alignment, all 6,936 validation appearances across the overlapping splits reproduced the saved predictions exactly. A separate reconstruction of the external cohort from source inputs also reproduced its original file digest. These checks establish computational reproducibility on this host; independent cross-platform replication remains untested.

The sequential five-seed reproduction took approximately 157 minutes wall-clock (29–35 minutes per seed), reconstructed from the supervisor start timestamp and per-seed completion-file timestamps. This includes repeated feature extraction and fitting all three branches, but excludes dependency installation and the broader research search; it is not an optimized fitting-only benchmark. Training used an Apple M4 Pro CPU with 24 GiB unified memory and six threads for the tree models, without paid cloud compute. Model complexity varies across runs: LightGBM contains 507–2,489 trees, CatBoost 1,143–2,200 trees, and SVR 4,284–5,852 support vectors. Saved model files occupy approximately 19–32 MB per run.

The dataset and benchmark protocol are available through AqSolDB and TDC [2, 3]. The research code is available at https://github.com/ruebenkadosh-ctrl/aqsol-blend, with frozen model files in release v0.1.0 at https://github.com/ruebenkadosh-ctrl/aqsol-blend/releases/tag/v0.1.0. The release records ordered feature definitions, model-file hashes, resolved dependencies, the external protocol, and the five-seed reproduction comparison.

6.2 Scope of the evidence

Development reused overlapping validation partitions for early stopping and model comparison. Four sequential evaluations were performed on the official test set, and earlier results and leaderboard information influenced subsequent research direction. Consequently, the final benchmark score is an adaptively exposed estimate. The ten-split comparisons are useful for assessing consistency within this development process, but do not provide independent confirmation. The external evaluation tests the frozen recipe on structurally screened molecules; a nested development evaluation of the final recipe has not been completed.

BigSolDB is a transfer dataset rather than an exact replication of the AqSolDB endpoint. Its records include differing measurement methods and frequently unspecified solid forms; pH is not supplied in the inspected metadata. Solvent-volume concentration conversion can differ from solution molarity at high concentrations. Exclusion of molecular overlaps does not eliminate shared literature practices, source-level dependence or other measurement biases. Molecule-level bootstrap intervals do not account fully for these effects.

The small difference from MiniMol cannot establish statistical superiority without an appropriate comparison. The reported standard deviations measure variability across runs, not uncertainty from adaptive model selection. Competitor predictions were unavailable for paired analysis. The solubility-range analyses likewise describe reused validation observations rather than independent subgroups of an untouched holdout.

Canonicalization checks found no identical molecules shared between each development validation partition and the remaining development rows. This check does not address all forms of chemical similarity or dataset overlap. Mordred descriptor failures are mapped to zero, including failures caused by its detour-calculation timeout. Hardware-dependent timeouts can affect extracted features. A subsequent label-free extraction audit recorded zero timeout-affected molecules among 1,997 TDC test inputs and 364 unique raw external SMILES (before parent aggregation and missing-target exclusion). Other undefined descriptor values occurred on every input: 383,761 descriptor cells on the test set and 78,735 externally, corresponding to 11.9% and 13.4% of the respective descriptor matrices. Recorded exceptions were ValueError, ZeroDivisionError and MultipleFragments. A spot check found absent atom-type extrema and undefined divisions among these cases. All follow the fixed zero-replacement policy. These counts describe the fresh audit, not historical timeout incidence, and do not establish stability across different hardware.

7 Conclusion

AqSol-Blend achieves a five-run TDC AqSolDB MAE of 0.736 ± 0.014 using a CPU-trained combination of LightGBM, CatBoost and SVR. The final Mordred substitution improves nine of ten development splits while preserving the other model branches and mixture weights. Changes to loss functions, seed averaging and tree capacity were less effective in the examined configurations. The external evaluation shows lower average error than a fixed random forest, while the smaller gain over the predecessor remains uncertain. Together, the experiments support the practical value of complementary representations and identify chemical novelty and measurement comparability as priorities for further validation.

AI assistance

OpenAI Codex was used as a supporting tool to locate sources, clarify research material and analysis outputs, assist with code development and refinement, and draft and revise manuscript text. The research ideas and direction were the author's, and the author takes responsibility for the work.

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